



Diagnostic Tests

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Objectives

At the end of this session, you will be able to:

- Identify the different diagnostic modalities used in clinical practice
- Differentiate abnormal lab values from normal (reference ones)
- Interpret different laboratory results of a patient
- Monitor patient response using laboratory values
- Identify drugs that will interfere with normal lab values
- Devise treatment strategy and develop pharmaceutical care plan based on patient lab investigations

Diagnostic Tests (General principles)

- Diagnostic test findings
 - should be used to **compliment** other **subjective and objective findings**.
 - and must not be evaluated in **isolation**.
- Diagnostic test findings can be helpful
 - in assessing clinical disorders
 - establishing a diagnosis,
 - evaluating disease progression and assessing drug therapy
- The findings must be assessed in the context of the clinical situation since different factors affect the test results.

Diagnostic Tests(General principles)

- Which factors affect lab values?
 - Patient factors, Lab factors, human factors
- Normal values may vary from laboratory to **laboratory, techniques and reagents**
- Drugs (masking)
 - Amitriptyline (blue green urine)
 - Rifampin (orange urine)
- Normal values may also vary depending on the **patient's**
 - **age (organ function); gender (Hgb), weight/height (SCr), and other factors.**

Diagnostic Tests(General principles)

- Laboratory error
 - technical error, improper calculation, inadequate specimen, incorrect sample timing, improper sample preservation, food or medication interference with laboratory tests.
 - If laboratory error is suspected, the test should be repeated.

Remember: ALWAYS Treat the Patient not the Lab Value!

Diagnostic Tests(General principles)

Reference Ranges vs Normal Range

- The term **reference range** is preferred in clinical practice
 - several factors contribute to the “normal” value for each individual.

Test Reliability

- Indicators of test reliability include **accuracy, precision, sensitivity, and specificity.**
- **Precision:** refers to the repeatability of a laboratory test
 - (i.e., test results fall within a similar value when repeated)
- **Accuracy:** the ability of a test to provide a result that is reflective of the “true” value
 - (i.e., the test result matches the actual real value).

Diagnostic Tests(General principles)

- Quality control and assurance practices at each laboratory are monitored regularly to ensure reliability of results.
- Typically, if a result is obtained that is significantly outside the reference range, the laboratory will repeat the test to confirm or refute the finding.
- Research studies generally establish the **sensitivity** and **specificity** of laboratory tests.
- Clinically these are essential to distinguish **the presence or absence of a disease or condition.**

Diagnostic Tests(General principles)

- **Sensitivity:** the ability of the test to correctly identify the disease or condition.
 - If a test is 95% sensitive, then 95% of the individuals will be correctly identified as having the disease or condition, but 5% will have a negative test result even though they have the disease or condition (false negative).
- **Specificity:** the ability of the test to rule out individuals who do not have the disease or condition.
 - If a test is 95% specific, then 95% of the individuals without disease will have a correct negative result, but 5% will be identified as having the disease or condition even though they are negative (false positive).

Diagnostic methods

A. Biochemical methods

Blood tests

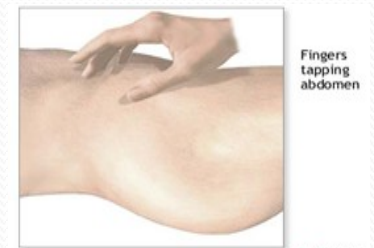
- Enzyme tests
- Test for evaluation of serum concentration - Na, K, Ca..., glucose, Urea
- Organ function test

Urine & other bodily fluid analysis

B. Physical methods

1. Mechanical

- Blood pressure
- Body temperature measurement
- Auscultation
- Palpation
- Percussion



Diagnostic methods

2. Electrical

- ECG, EEG, Audiometry ...

2. Electromechanical

- Spirometry

4. Optic and Optoelectric

- Light microscopy, Electron microscopy, Ophtalmoscopy, Otoscopy, Bronchoscopy...

5. Ultrasound (Dopplers)

- Blood flow test, Sonography, Echocardiography...

6. X- ray Imaging

- Computer tomo-graphy (CT), CXR.....

7. Methods of Nuclear Medicine

- Radioi-sotopes , Gammagraphy, Positron Emmision Tomography (PET)....

8. Magnetic scanning

- Nuclear Magnetic Resonance Tomography (NMRT)

Diagnostic Tests- Microbiology

Microbiologic Studies

- Direct examination of a specimen (eg, sputum, blood, or urine) may aid in a presumptive diagnosis.
- Microbial cultures are obtained with a Gram stain of the cultured material.
- A Gram stain of collected specimens **can give rapid information** that can be applied immediately to patient care.

Gram Stain

- Developed in 19th Century for distinguish and classify bacterial species.

Diagnostic Tests- Microbiology

Gram Stain

- **Three classifications** – color; shape & organization
- **Color** - Two classes of bacteria
 - Gm +ve – stain purple
 - Gm –ve stain pink or red
 - Bacteria indwelling intracellularly...?
 - Chlamydia, Mycobacterium (without cell wall)
 - identified by alternative staining techniques
- **Shape** – Cocci (round); Bacilli (rod)
- **Organization** – Pairs or clusters; chains

Gram Stain

Positive

Cocci

CLUSTERS Staphylococci

Coagulase-positive
Staphylococcus aureus

Coagulase-negative
Staphylococcus epidermidis
Staphylococcus saprophyticus
Staphylococcus hominis
Staphylococcus hemolyticus
Staphylococcus warneri

PAIRS (Diplococci)

Pneumococci
Streptococcus pneumoniae

CHAINS Streptococci

β-Hemolytic
Streptococcus pyogenes
Streptococcus agalactiae
Groups C, F, G

α-Hemolytic
Viridans *Streptococcus*
Streptococcus pneumoniae

Bacilli

SMALL

Listeria
Propionibacterium
Corynebacterium
Gardnerella

LARGE

Spore-forming
Clostridium
Bacillus

Nonspore-forming
Lactobacillus

BRANCHING or FILAMENTOUS

Nocardia
Actinomyces
Erysipelothrix

Negative

Cocci

Neisseria meningitidis
Neisseria gonorrhoeae
Veillonella

Bacilli

LACTOSE FERMENTER Oxidase-positive

Aeromonas
Pasteurella
Vibrio

Oxidase-negative

Escherichia coli
Klebsiella spp.
Enterobacter spp.
Citrobacter spp.

NONLACTOSE FERMENTER

Oxidase-positive
Pseudomonas spp.
Flavobacterium spp.
Alcaligenes spp.
Achromobacter spp.
Moraxella spp.

Oxidase-negative

Proteus spp.
Providencia spp.
Serratia spp.
Morganella spp.
Salmonella spp.
Shigella spp.
Stenotrophomonas
Acinetobacter spp.
(coccobacilli)

Culture and susceptibility

- Culture and susceptibility testing provides additional information to the clinician to select appropriate therapy.
- Specimens are placed in or on culture media that provide proper growth conditions.
- Once the bacteria grow on culture media, they can be identified through biochemical tests.
- When a pathogen is identified, susceptibility tests can be performed to various antimicrobial agents.
 - Reported as: Sensitive (S) vs Intermediate (I) vs Resistant (R)



Diagnostic Tests: Hematology

Case

- ❖ **R.L.**, a 45-year-old man, is hospitalized with a sustained high fever of 39.4 °C, SOB.
- ❖ His cough is productive of rusty sputum, and he appears to be in acute distress.
- ❖ The results of the CBC and leukocyte differential are as follows:
 1. **Total WBC count-**
18,000/ μ L

WBC Differentials

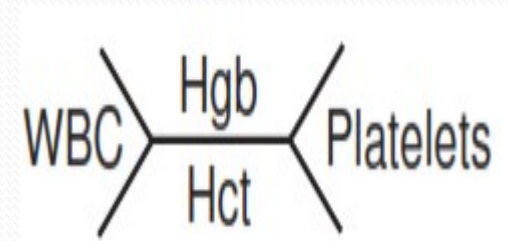
1. **Neutrophils-76%**
 2. **Bands- 13%**
 3. **Lymphocytes- 10%**
 4. **Monocytes- 0**
 4. **Eosinophils- 1%**
 4. **Basophils- 0**
- On the basis of this laboratory report and other findings, a diagnosis of *pneumococcal pneumonia* is suspected.
- ☞ How **R.L.**'s laboratory report consistent with bacterial infection?

HEMATOLOGY

- Hematology involves the diagnosis and treatment of diseases of the blood and includes the examination of peripheral blood and the blood precursor cells found in the bone marrow.
- There are several different hematologic cell types that originate from the hematopoietic stem cell.
- Each cell line has a defined role and unique contribution to the overall homeostatic process, and may be found in the **bone marrow, lymph system, or blood.**
- Typically, routine clinical laboratory testing involves **measuring concentrations of mature myeloid cells found in the blood.**

Complete blood count (CBC)

- CBC is one of **the most commonly ordered clinical laboratory tests**.
- A CBC measures the red blood cells (RBCs), hemoglobin (Hgb), hematocrit (Hct), mean cell volume (MCV), mean cell Hgb concentration (MCHC), and total white blood cells (WBCs).
- Depending on the laboratory, CBC may also include platelets, reticulocytes, or leukocyte differential.
- An abbreviated method of noting hematologic parameters in clinical practice is noted in the following figure.

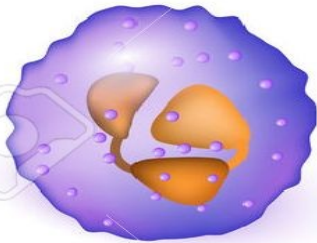


White Blood Cells ($4-11 \times 10^3/\mu\text{L}$)

WHITE BLOOD CELL

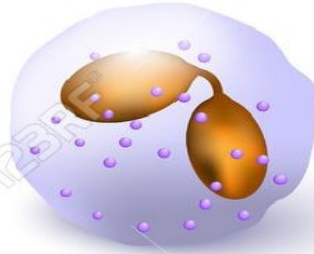
Granulocytes

Neutrophil



(phagocytosing a bacteria and other pathogens)

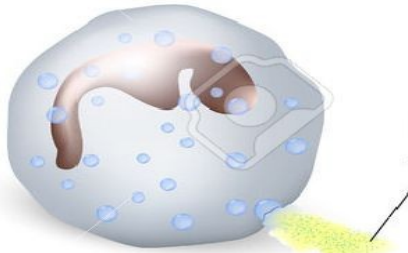
Eosinophil



(control mechanisms associated with allergy)

Basophil

(contain histamine and heparin)



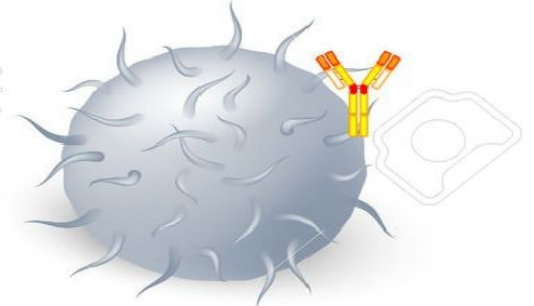
Histamine release from the basophils

Agranulocytes

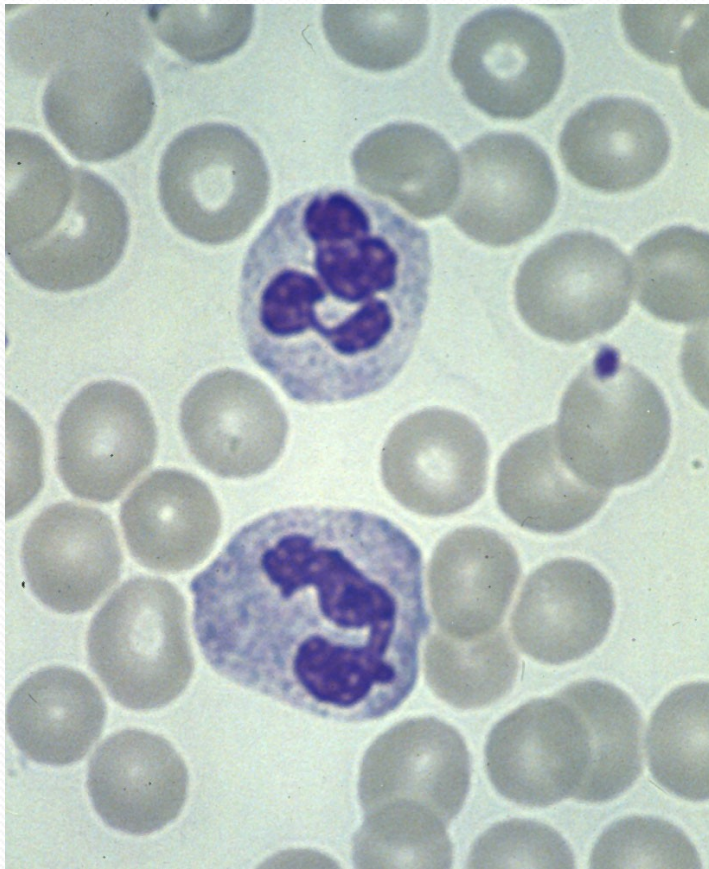
Monocyte (phagocytosis)



Lymphocyte (secretion of antibodies)



The neutrophil (40–70% of WBC)



- Segmented neutrophil
- *Life span of about 10 days*
- Moves from bone marrow to blood to tissues
- Mature more quickly **under stressful conditions**
- **Granulocytes**
- **Primary defense** against **bacterial infection**
 - **Locate, ingest, and kill bacteria and other foreign invaders**

The neutrophil: neutrophilia

➤ Increase in neutrophil count

1. Pathologic

- Bacterial infection
- Certain viruses and fungi
- Inflammatory responses to tissue death
 - Burns
 - Snake bites

2. Drugs

- Steroids
- Lithium-increase marrow input

3. Increased Production of WBCs

4. Physiologic

- Pseudo-neutrophilia shift of cells
 - Catecholamine
 - ® Acute stress

5. Other inflammatory responses

- Neoplastic growth
- Metabolic disorders

The neutrophil: neutropenia

1. Decreased Production of WBCs

- Bone marrow diseases
- Malignancies that affect the bone marrow

2. Increased Neutrophil Destruction

- Overwhelming infection
- Certain bacteria
- Immune reactions

3. Pseudo-neutropenia

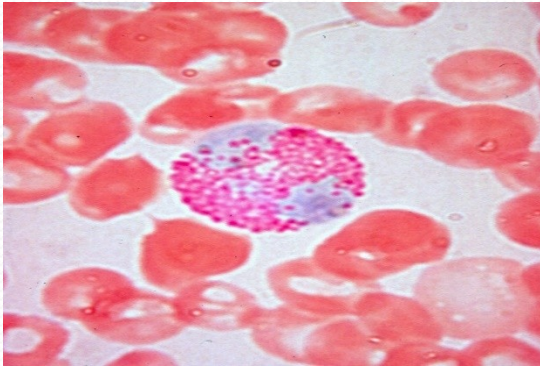
- Viral infections
- Hypothermia-marigination

Agranulocytosis & Absolute Neutrophil Count

- Degree of Neutropenia:
ANC < 1500 (cells/ μ L.)
- **Agranulocytosis severe neutropenia (ANC < 500 cells/ μ L.)**
- The risk of developing **bacteremia** is increased further as the ANC decreases to <100/ μ L called
 - “**profound neutropenia**”

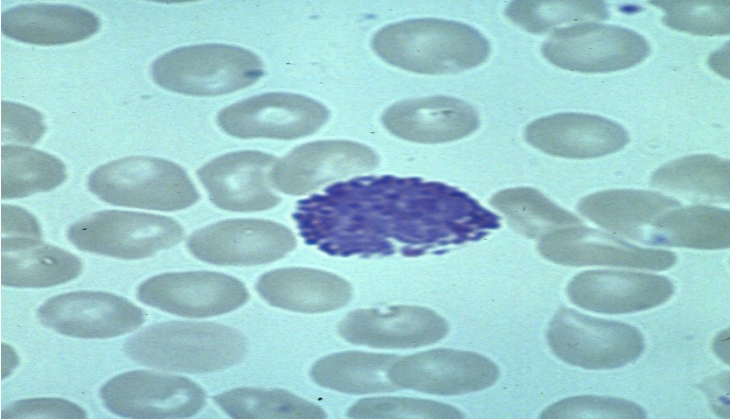
Infection Risk	ANC
Mild	≥ 1000 to < 1500 cells/ μ L
Moderate	≥ 500 to < 1000 cells/ μ L
Severe	< 500 cells/ μ L

The eosinophil (0–8% of WBC)



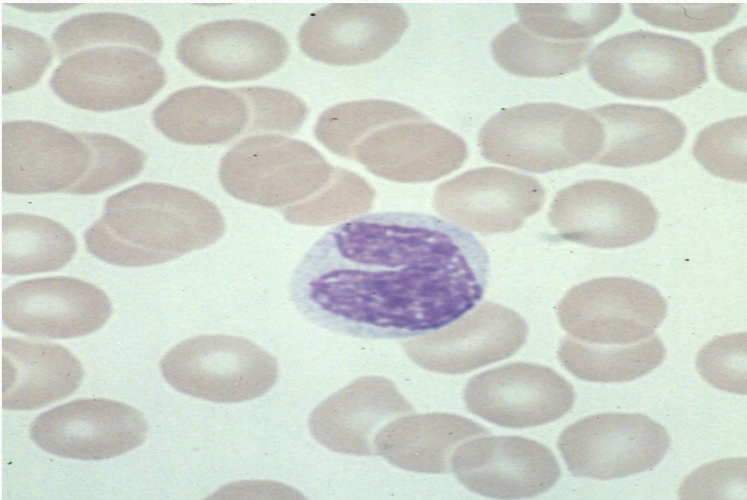
- Segmented eosinophil
- **Life span = 14 days**
- Spends little time in the blood before it locates in the **skin, GI tract,** or **respiratory** tract
- Only 1% of mature cells are located in blood
- **Phagocytic role**
 - ® But appear to be **less potent** than neutrophil
- Drawn to sites of **hypersensitivity reactions/Allergic/** by mast cell chemotactic factors
- Found in sputum of **asthmatics**
 - ® May play a role in **pathogenesis of lung disease**
- Play a role in **parasitic infections**

The basophils(0–3% of WBC)



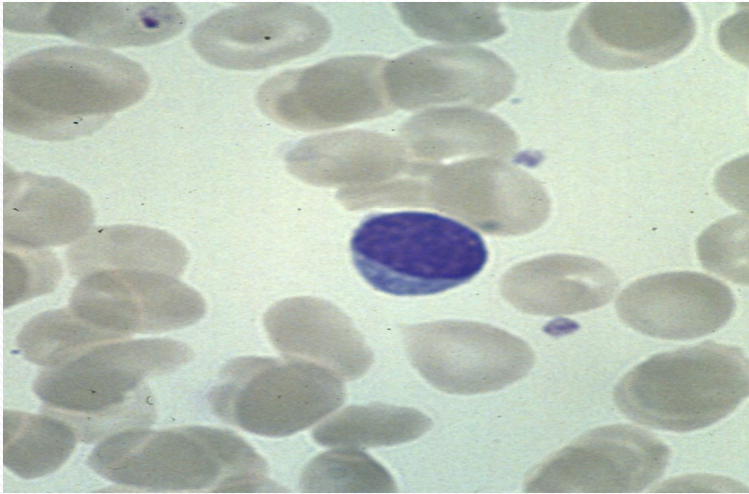
- **Least common** type of **WBC**
- Found in tissues in which inflammation
 - ® Resulting from **hypersensitivity** to proteins, contact allergy.
- Increase in response to same conditions that cause **eosinophils** to respond
- Can phagocytose sensitized red cells but are less active phagocytes than the other granulocytes.
- Lack significant amounts of antibacterial or lysosomal enzymes.

The monocytes (0–11% of WBC)



- Also not common in **circulating blood**
- Stay in blood for about 70 hours
- Become **macrophages** in tissue and live for several months or longer
- Primary role is **phagocytosis**
 - Play large role in ingesting cellular debris
- “**Activated**” when direct contact with microorganisms
 - Has greater motility, enzyme activity and killing capacity (causes fever)
- Also play a role in **immunity**

The lymphocytes (20–40% of WBC)



- Can pass back and forth between blood and tissues
- **Two types**
 1. **B-lymphocytes**
 2. **T-lymphocytes**
- Main function is **antigen recognition and immune response**
- Life span quite varied (up to **two years**)

Lymphocytes: B types

- B cells are not only produced in the **bone marrow** but also mature there.
- **B lymphocytes (or B cells)** are most effective against **bacteria & their toxins** plus a few **viruses**
- Once **B cells** encounter their antigen and receive a cytokine signal from **helper T cells**, differentiate into **plasma B cells** and **memory B cells**

Lymphocytes: B types

1. **Plasma B cells** secrete **antibodies** that induce the destruction of target antigens
 2. **Memory B cells** play an important role in **long-term immunity**
- B cells play a large role in the **humoral immune response**

Lymphocytes: T types

- Precursors of T cells leave the **bone marrow** and mature in the **thymus**
- Role in **cell-mediated immune** response
 - **Recognize & destroy abnormal body cells,**
 - **Including virus-infected cells & cancer cells.**
- There are 3 subclasses of T cells
 - 1. Cytotoxic T cells (CD8+)**
 - **Promote target-cell destruction by**
 - I. Activating cellular apoptosis or**
 - II. Aggressively killing the target cell via the release of cytotoxic proteins**

Lymphocytes: T types

2. Helper T cells (CD4+)

- The great communicators of the immune response
 - ✓ Proliferate and secrete cytokines that regulate effector cell function
 - ✓ Some secrete cytokines that recruit **cytotoxic T cells, B cells**

2. Regulatory T cells or suppressor T cells

- ✓ Suppress the activation of an immune response

Lymphocyte count

I. Decreased

- A. AIDS
- B. Bone marrow suppression
- C. Aplastic Anemia
- D. Steroids
- E. Neurologic Disorders
 - 1. Multiple sclerosis
 - 2. Myasthenia gravis
 - 3. Gullain barre syndrome

II. Increased

- A. Influenza
- B. Tuberculosis
- C. Mumps
- D. Cytomegalovirus Infection
- E. Infectious Hepatitis
- F. Viral pneumonia

Interpreting the CBC

- What is the total white cell count?
 - ❖ **WBC: $4 \times 10^3 - 11 \times 10^3$ cells /uL**
- If elevated ($>11,000$), what type of WBC is the culprit?
 - Is it the **neutrophils**, eosinophils, lymphocytes, basophils, or monocytes?
- Marked **leukocytosis** is usually due to **neutrophils or lymphocytes**.

Interpreting the CBC

- If the **neutrophils** are causing the **leukocytosis**, compare the neutrophil % to **total WBC**.
 - The % neutrophils indicates the **severity of the infection**
 - The **total WBC** reflects the **quality of the immune system**

❖ Example (Case # 1)

- An 85 years old female with pneumonia:
 - ✓ **Total WBC is: 11,500**
 - ✓ Neutrophil % = 80% (9200)
 - ✓ Bands = 5%
- Indicates that a **severe infection** is present but the immune system cannot **respond appropriately**.
 - Prognosis poor

Interpreting the CBC (Case # 2)

- A 5-year-old male with pneumonia
 - ✓ **WBC = 18,000**
 - ✓ Neutrophils = 60% (10,800)
- **Marked leukocytosis and normal range for neutrophils indicates**
 - **Mild to moderate infection but excellent immune system response**

❖ **Excellent prognosis**

Interpreting the CBC (Case #3)

- A 10-year-old male admitted for pneumonia
- WBC: 16,000 **Reference Range**

Neutrophils = 75% (12,000) (1800-7500)

Bands = 5% (800) (0-100)

Eosinophil = 1% (160) (0-600)

Lymphocytes = 10% (1600) (900-4500)

Basophils = 0% (0) (0-100)

Monocytes = 3% (480) (90-1000)

Interpreting the CBC (Case #3)

❖ Interpretation

- Neutrophilia probably due to bacterial pneumonia
- Left shift indicative of severe infection
- The **source** of the **neutrophils** is the **bone marrow** since many **bands** are present

Interpreting the CBC (Case #4)

- A 20-year-old male admitted following **Motor Vehicle Accident**
 - ✓ WBC -**14,500**
 - ✓ Neutrophils- **75%**
 - ✓ Bands- **1%**
- ❖ **Leukocytosis** due to neutrophilia
- ❖ **History** and **low percent of bands** suggest pseudo-neutrophilia
 - **Due to the liberation of marginated neutrophils in the intravascular system**

Platelet (thrombocytes)

- Platelets, in combination with several other factors such as blood vessel wall and coagulation factors, play key role in hemostasis.
- Platelet count: $150\text{--}450 \times 10^3/\text{mm}^3$ ($150\text{--}450 \times 10^9/\text{L}$)
 - Thrombocytopenia $<150 \times 10^3/\text{mm}^3$
 - Bleeding $<50 \times 10^3/\text{mm}^3$
 - Thrombocytosis $>450 \times 10^3/\text{mm}^3$
- Thrombopoietin, a hormone synthesized in the liver, regulates platelet production



Life span of about 10days

Platelet (thrombocytes)

❖ Thrombocytosis

- Acute infection
- Chronic leukaemia
- Inflammation
- Heart disease
- Iron deficiency anemia
- Polycythemia
- Rheumatoid arthritis
- Sickle cell anemia
- Trauma
- Tuberculosis

❖ Thrombocytopenia

- AIDS
- Hemolytic anemia
- Idiopathic Thrombocytopenic Purpura
- Menstruation
- Radiation
- Splenomegaly
- Pernicious anemia
- Prosthetic heart valve

Red Blood Cells



- Produced in bone marrow
- Life span of about **120days**
- Primary function is **oxygen** transport
- Immature forms have nucleus and called **reticulocytes**

Red Blood Cells

- Most numerous among blood cells
- **Lack nuclei** and contain **hemoglobin**, iron-containing protein that acts in the transport of oxygen and carbon dioxide
- Defined by three quantitative values
 1. The red cell concentration per unit volume
 2. The amount of hemoglobin (Hgb.)
 3. The volume of packed red cells or hematocrit (Hct.)

Red Blood Cells

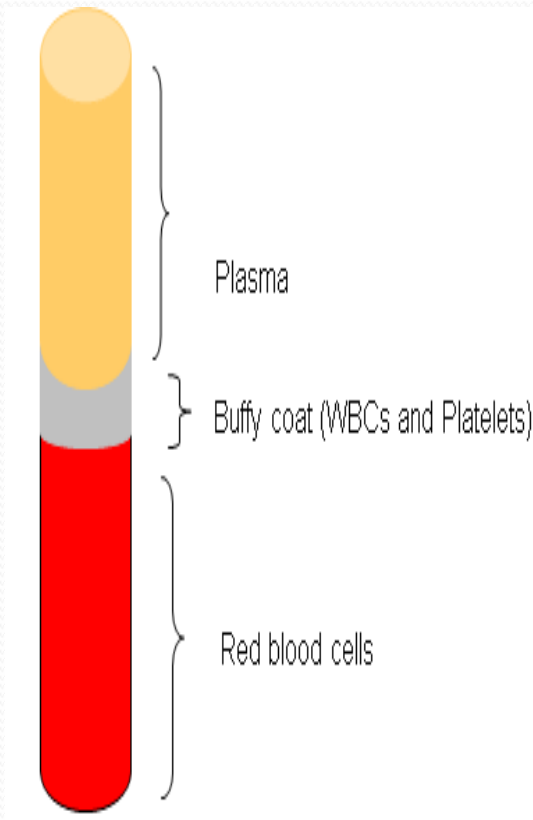
1. Normal values:

Males: $4.3\text{--}5.9 \times 10^6 /\mu\text{L}$ or $4.3\text{--}5.9 \times 10^{12}/\text{L}$

Females— $3.5\text{--}5.0 \times 10^6 /\mu\text{L}$ or $3.5\text{--}5.0 \times 10^{12}/\text{L}$

Anemia

- Abnormal *Decrease* in RBC count
- RBC counts are decreased in various types of anemias, lymphomas, and leukemia.
- ▣ After puberty, females have lower RBCs and Hgb. due to menstrual bleeding.



Red Blood Cells

Increased RBC count = Polycythemia

- A. Primary- **bone marrow** problem, cancer
- B. Secondary
 - ✓ **Living at high altitude**
 - ✓ **Chronic lung/heart disease**
 - ✓ **Tobacco use/carbon monoxide**
- C. Relative Polycythemia
 - ✓ **Dehydration**

Red Blood Cells

❖ Increased

- Cardiovascular disease
- Chronic hypoxia
- Chronic lung disease
- Congenital heart defect
- Hepatic cancer
- High altitude
- Polycythemia vera
- Smoking

❖ Decreased

- Alcohol abuse
- Anemias
- Bone marrow suppression
- Chronic infection
- Chronic renal failure
- Hemodilution
- Hemorrhage
- Hypothyroidism
- Rheumatic fever
- Systemic lupus erythematosus
- Vitamin deficiency (B6, B12 , folic acid)

Reticulocyte count

Ⓜ Normal values:

- **0.5 – 1.5% of RBC**

❖ Role

- To identify cause of *anemia*
- Increase indicates *anemia* is due to *blood loss*
- Decrease indicates *anemia* is due to *bone marrow disease*
 - Infectious disease, CKD, untreated Iron Deficiency Anemia, alcoholism

Reticulocyte Count

- **Increased**

- Effective treatment of anemia
- Hemolytic anemia
- Sickle cell anemia

- **Decreased**

- Cancer
- Chronic infection
- Cirrhosis of the liver
- Folate deficiency
- Iron deficiency anemia
- Pernicious anemia
- Radiation therapy
- Alcoholism
- Aplastic anemia
- Bone marrow depression

Hemoglobin (Hgb.)

- Hgb. is the oxygen-carrying compound contained in RBCs.
 - *Males: 14-18 g/dL*
 - *Females: 12-16 g/dL*
- Decreased Hgb.
 - Anemia
 - Fluid replacement
- Increased Hgb.
 - Polycythemia Vera
 - COPD
 - Chronic smokers
 - Vigorous exercise
 - Living at high altitude

Hematocrit(Hct)

- The volume of blood that is occupied by RBCs.

- ▣ It is expressed as a percentage of total blood volume.

- Male: 39%-49%

- Female: 33%-43%

- Decreased

- ▣ Anemia

- ▣ Leukemia

- **Increased**

- ▣ Hemo-concentration

- ▣ Polycythemia vera

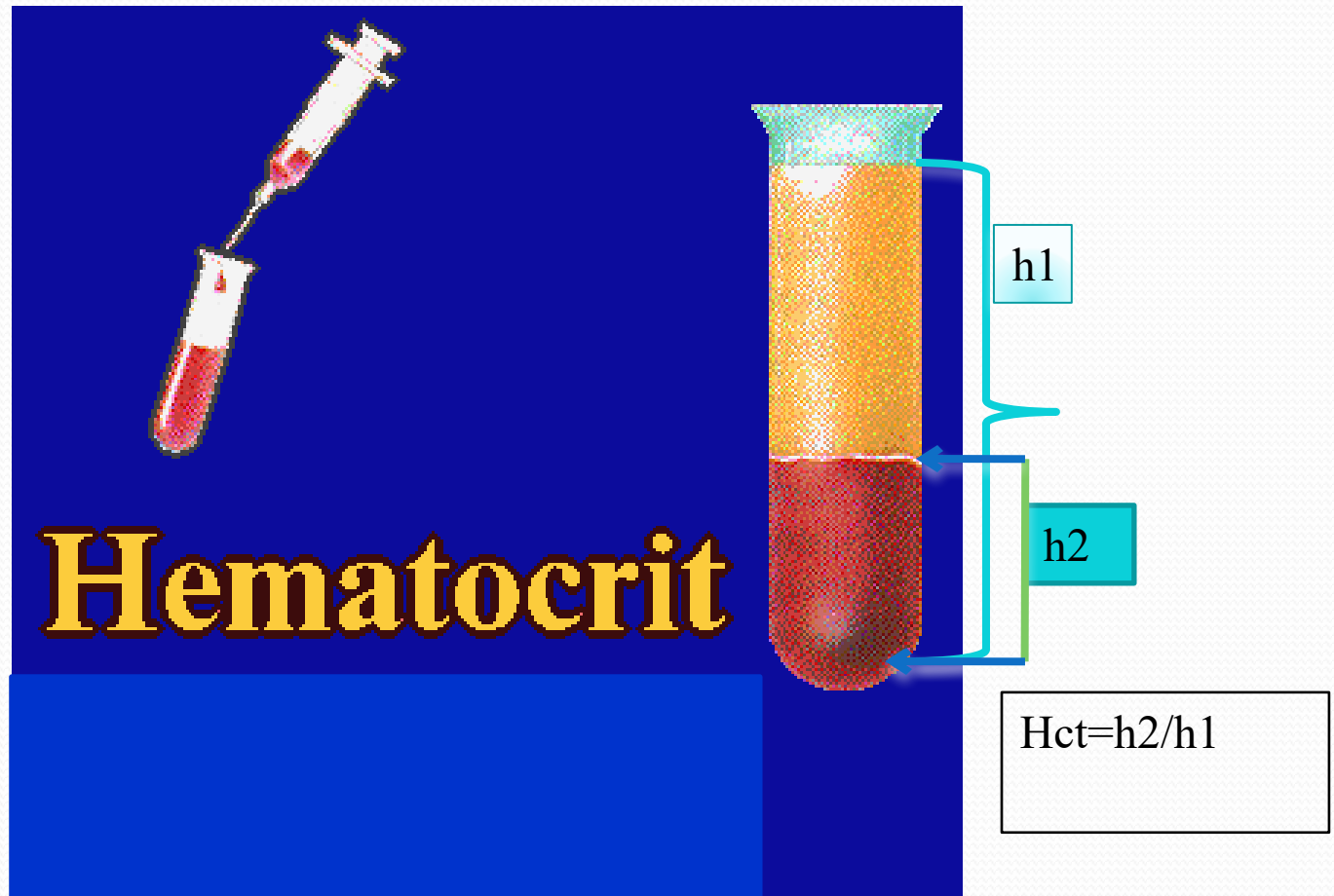
- ▣ COPD

- ▣ High altitude

- ❖ **Determined**

- By centrifuging a capillary tube of whole blood and comparing the height of the settled RBCs to the height of the column of whole blood.

Hematocrit(Hct)



The rule of three

- Applies to normocytic, normochromic erythrocytes only
- Useful **to detect laboratory error** in measuring the Hgb, Hct, and RBC count
 - $\text{Hct} = 3 \text{ times Hb}$
 - $\text{Hgb} = 3 \text{ times the RBC count}$

❖ Example

- $\text{RBC} = 3.0 \times 10^{12}$; $3 \times 3 = 9$
- $\text{Hgb} = 9.2 \text{ g/dL}$ 3
 $\times 9.2 = 27.6$
- $\text{Hct} = 28\%$

® **Is the rule of 3 working?**

Red blood cell indices

- Describe average **qualitative characteristics of the red cell population**
 - RBC indices
 - Useful in the classification of anemia.
 - Includes
 1. MCHC
 2. MCV
 3. MCH
1. **MCHC (33-37g/dl)**
 - Measure of the concentration of hemoglobin in an average RBC

$$\text{MCHC} = \text{Hgb} / \text{Hct}$$

- I. ***Hypochromic*** anemia
 - IDA, Hypothyroidism
- II. ***Normochromic*** anemia.
 - Recent blood loss, Hemolysis
 - Chronic disease
- III. **Hyperchromic** anemia
 - Sick cell

Red blood cell indices

2. MCV (80-100 μm^3)

- Volume occupied by single RBC

$$\text{MCV} = \text{Hct} / \text{RBC count}$$

- **Microcytic anemia** (<80 fL)
 - Iron deficiency
- **Normocytic anemia**
 - Blood loss
 - Hemolytic anemia

- **Macrocytic anemia** (>100 fL(μm^3))

- Folic acid deficiency
- Vitamin B12 deficiency
- habitual alcohol ingestion
- chronic liver disease,
- Some COPD patients

3. MCH (27-33pg/cell)

- Measures average weight of Hgb in RBC. Less reliable than MCHC.

$$\text{MCH} = \text{Hgb} / \text{RBC}$$

Interpreting the red blood cells

CBC:	Results		Reference range
RBC ($\times 10^{12}/L$)	3.5		3.5-5.0
Hgb (g/dL)	10.6	↓	11.5-15.5
Hct	34.9%	↓	39%-49%
MCV (μm^3)	77.0	↓	80-100
MCHC (g/dl)	30.4	↓	33-37

Interpretation: Microcytic, hypochromic anemia; rule of 3 does not apply

Erythrocyte sedimentation rate (ESR)

- The rate at which erythrocytes **settle** to the bottom of a test tube through the forces of **gravity** and in response to **fibrinogen levels** in the blood.
 - ❖ Males: 0-20 mm/hour
 - ❖ Females: 0-30 mm/hour
- **Nonspecific value** and may be increased abnormally in
 - acute and chronic **inflammatory processes**,
 - acute and **chronic infections**
 - neoplasms
 - infarction
 - tissue necrosis,
 - rheumatoid-collagen disease
 - nephritis

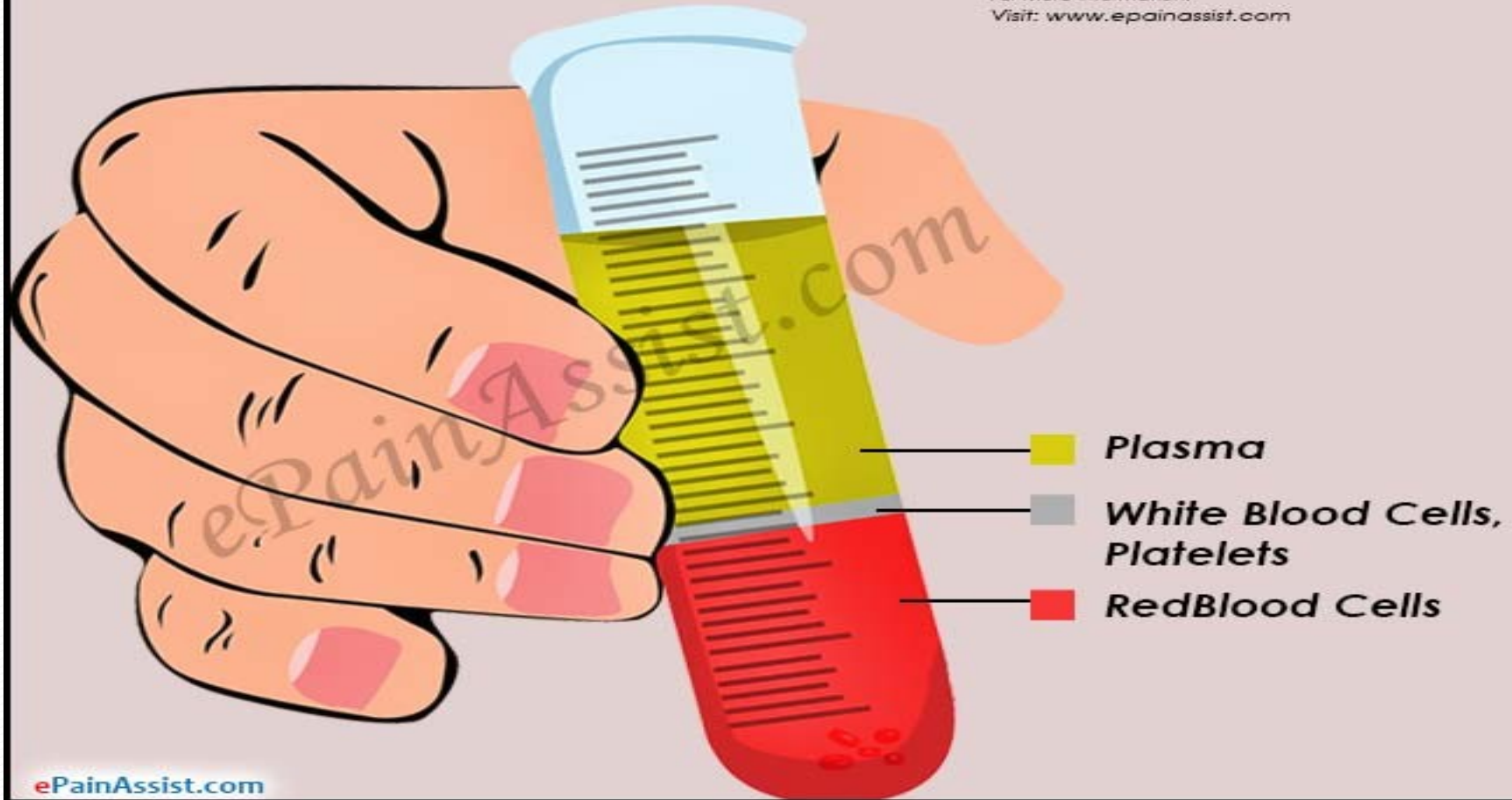
ESR

What Happens if Erythrocyte Sedimentation Rate is High?

The higher the Erythrocyte Sedimentation Rate or ESR, the more likely are the chances that there is inflammation present in the body.

For More Information:

Visit: www.epainassist.com



ESR

❖ Increase

- Anemia
- Infection
- Inflammatory process
- Myocardial infarction
- Osteomyelitis
- Rheumatoid arthritis
- Tissue injury
- Systemic lupus erythematosus

❖ Decrease

- Congestive heart failure
- Polycythaemia Vera
- Sickle cell anemia

Coagulation Studies

- ❖ The **prothrombin time (PT)**, **international normalized ratio (INR)** and **activated partial thromboplastin time (aPTT)** are used:
 - **to diagnose coagulation abnormalities or**
 - **to monitor the effectiveness of patients receiving anticoagulation therapy.**
- When used to assess drug therapy, achieving a value outside the reference range is a therapeutically desirable outcome.

Coagulation Studies

PROTHROMBIN TIME (PT)

Reference Range: 10–14 seconds

- Prothrombin is synthesized in the liver and is converted to thrombin during the blood clotting process.
- Thrombin formation is the critical event in the hemostatic process because thrombin creates fibrin monomers that ultimately assemble into a clot and stimulates platelet activation.
- The PT test directly measures the activity of clotting factors **VII, X, prothrombin (factor II), and fibrinogen.**
- Automated laboratory instruments measure PT by recording the time required for the blood to clot after tissue thromboplastin and calcium has been added to the patient's blood sample

Coagulation Studies

ACTIVATED PARTIAL THROMBOPLASTIN TIME (aPTT)

Reference Range: 20–39 seconds

- aPTT measures the **intrinsic clotting system**, which depends on factors VIII, IX, XI, and XII and the factors involved in the **final common pathway** of the clotting cascade (factors II, X, and V).
- **aPTT is commonly used to monitor unfractionated heparin therapy.**
- An aPTT below the desired therapeutic range indicates the need to rebolus and/or increase the heparin infusion rate.
- An aPTT above the desired therapeutic indicates the need to hold and/or reduce the dose of heparin....risk of bleeding

Coagulation Studies

INTERNATIONAL NORMALIZED RATIO (INR)

(Usually, INR= 2-3)

- *For initiation & maintenance of anticoagulant therapy (e.g. Warfarin).*
- $$\text{INR} = \frac{\text{PT (patient)}^{\text{ISI}}}{\text{PT (control)}^{\text{ISI}}} = \text{PTR}^{\text{ISI}}$$
- PTR =the prothrombin ratio.
- ISI= international sensitivity index.
- Commercial manufacturers quantify the ISI for the specific thromboplastin reagent used in each lot and report this information in the product package insert.
- An INR below the desired range indicates *suboptimal anticoagulation and a need to increase warfarin dosage.*
- Conversely, an INR above the desired range indicates *a need to omit and/or reduce the warfarin dosage...risk of bleeding*

Blood chemistry

Blood chemistry

- ❖ Routinely used test
- Quickly provide basic information concerning **organ function** at a relatively low cost
- May vary among clinical laboratories.

Blood chemistry: Common clinical laboratory panels

Arthritis Panel (ARTH PN)

Uric acid, ESR, ANA (antinuclear antibody screen), rheumatoid factor

Basic Metabolic Panel (BC MET)

Creatinine, CO₂, chloride, glucose, potassium, sodium, BUN, calcium

Comprehensive Metabolic Panel (CM MET)

Albumin, alkaline phosphatase, ALT, AST, total bilirubin, calcium, CO₂, chloride, creatinine, glucose, potassium, sodium, total protein, BUN

Electrolytes (LYTES)

CO₂, chloride, potassium, sodium

Hepatic Function Panel (HEPFUN)

ALT, albumin, alkaline phosphatase, AST, direct bilirubin and total bilirubin, total protein

Acute Hepatitis Panel (ACUTE HEP)

Hepatitis A, AB, IgM, hepatitis B core antibody, IgM, hepatitis B surface antigen, IgM, hepatitis C, AB

Lipid Panel (LIPID PN)

Cholesterol, HDL, triglycerides (LDL and CHO/HDL ratio included, as calculated values)

Obstetric Panel (OB PN)

Type and Rh, antibody screen, RPR, rubella Ab-IgG, hepatitis B surface antigen

Prenatal Screen (PRESCP)

Type and RH, antibody screening and studies if indicated, RPR for syphilis, rubella Ab-IgG, hepatitis B surface antigen



Glucose

Blood Glucose (70–99 mg/dL : fasting)

- Blood glucose regulation is achieved through a complex set of mechanisms that involves insulin, glucagon, cortisol, epinephrine, and other hormones.
- Insulin and glucagon play a critical role in controlling blood glucose
- Because plasma glucose concentrations fluctuate in response to ingestion of meals, most glucose concentrations are **measured in either the fasting state or the post-prandial state**, depending on the type of information desired.
- Generally, normal glucose values refer to the plasma glucose concentration **in the fasting state**.

Blood Glucose

Glycosylated Hemoglobin (HbA1c: 4%–5.6%)

- Hgb is the oxygen-carrying component of the RBC.
- During the functional life span of RBCs (~4 months), glucose molecules bind to Hgb, which results in glycosylated Hgb (A1c).
- The concentration of Hgb A1c reflects a patient's average blood glucose concentration for the life span of circulating RBCs.
- In contrast, fasting glucose serum concentrations can fluctuate acutely based on either meals or insulin use.
- As a result, measurement of Hgb A1c concentrations provides a much better tool for evaluating chronic diabetes therapy.
- In a patient without diabetes, about 5% of Hgb is glycosylated; however, poorly controlled patients with diabetes can have Hgb A1c values greatly exceeding 8%.

Blood Glucose

- **HYPERGLYCEMIA**

- FBS > 126 mg/dl
- HgA1c > 6.5%
- Random blood sugar > 200 mg/dL

- Cause:

- Diabetes mellitus is the most common cause of hyperglycemia.
- Cushing's disease
- Shock
- Drugs: e.g Steroids, Atypical Antipsychotics

- Note: FBS 100 to 125g/dl → Impaired fasting glucose ...suggestive of pre-diabetes

- **HYPOGLYCEMIA**

- Blood glucose < 60mg/dl

- Cause

- Insufficient carbohydrate intake
- Addison disease
- Antidiabetic therapy



Lipid Profile

Lipid Profile

- Lipids are produced in liver.
- From liver lipids are not able to move to the bloodstream because they are insoluble in blood plasma
- So, liver wraps protein around the lipid resulting in new molecule called lipoprotein
- Lipoprotein move to blood stream throughout the body
- There are 3 major lipids in our blood
 - **Cholesterol:**
 - necessary for the synthesis of bile acid
 - **Triglycerides:**
 - provides energy to cells
 - **Phospholipids**
 - component of cell membrane

Total Cholesterol (<200 mg/dL)

- Total cholesterol testing evaluates
 1. The **risk for atherosclerosis**, myocardial occlusion, and coronary arterial occlusion.
 2. **Part of thyroid function**, liver function, renal function, and diabetes mellitus studies.
 3. Used to monitor **effectiveness of diet, medications**, lifestyle changes (e.g., exercise), and stress management.

Total Cholesterol (<200mg/dL)

❖ Increased

1. Cholestasis
2. Nephrotic syndrome, glomerulonephritis
3. Chronic renal failure
4. Hypothyroidism
5. Poorly controlled diabetes mellitus
6. Alcoholism
7. Diet high in cholesterol and fats (“dietary affluence”)
8. Obesity
9. Drugs

Total Cholesterol (<200mg/dL)

❖ Decreased

1. Severe hepatocellular disease
2. Hyperthyroidism
3. Malabsorption syndrome, malnutrition
4. Severe burns, inflammation
5. Conditions of acute illness, infection
6. Lipid-lowering drugs

Triglyceride (TG <150 mg/dL)

- Insoluble in plasma ,hence carried in lipoproteins to various tissues for energy utilization
- Associated with cardiovascular events
- ↑TG and LDL or ↓ HDL are risk factors for cardiovascular disease.

Types of lipoprotein

1. Very low-density lipoprotein (VLDL)

2. Low-density lipoprotein (LDL)

3. High-density lipoprotein (HDL)

- Firstly, the liver releases VLDL in the bloodstream.
- VLDL provides triglyceride to various cells for function

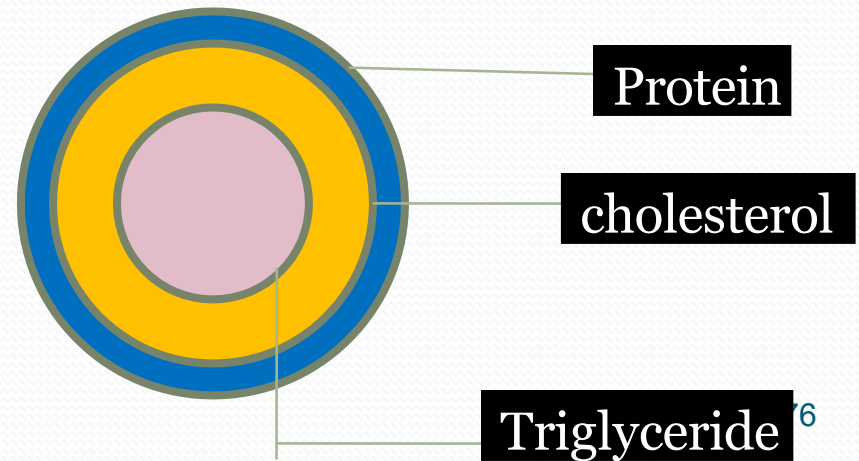
- After utilizing triglyceride, VLDL becomes LDL.
- LDL provides cholesterol to various cells
- If LDL production is increased, it will be deposited to the walls of artery making **plaque**.
- This plaque narrows blood vessels.....**bad cholesterol**
- Lipoproteins are measured in the blood of fasting (10– 12 hours) patients

Types of lipoprotein

- Liver also makes HDL that removes excess cholesterol from cells and plaque , and returns to liver **good cholesterol**

VLDL

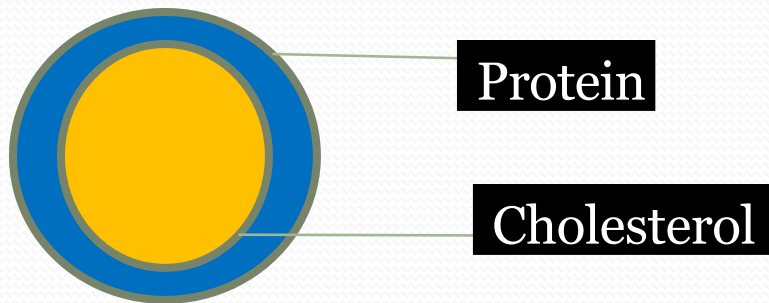
- composed of low level of protein,& high level of cholesterol & triglyceride



Lipoproteins

LDL

- composed of only protein and cholesterol
-



❖ LDL –C (<100 mg/dL)

- 60 to 70% of the total serum cholesterol
- Longer half-life (3 to 4 days) than its precursor VLDL, hence more prevalent in the blood
- “**Bad cholesterol**”

LDL-C (<100 mg/dL)

- Half removed from the systemic circulation by the liver
 - The other half may be taken up by peripheral cells or deposited in the intimal space of
 1. Coronary
 2. Carotid
 3. Peripheral arteries
 - Where **atherosclerosis** can develop.
- This is why LDL-C is the primary target of cholesterol-lowering therapy
 - The probability that atherosclerosis will develop is directly related to
 1. The **concentration** of LDL-C in the systemic circulation
 2. The **length of time** this level of **exposure** persists

$$\text{LDL cholesterol} = \text{total cholesterol} - \text{HDL cholesterol} - \frac{(\text{triglycerides})}{5}$$

HDL-C

Reference $\geq 40\text{mg/dL}$ (at risk: Men: <40 & women <50)

- It composed of more amount of protein and very less amount of cholesterol
- Reversely transport cholesterol from peripheral cells (i.e., lipid-rich inflammatory cells in the arterial wall) back to the liver
- Hence “good cholesterol”

